Autoencoder-based Continual Outlier Correlation Detection for Real-Time Traffic Flow Prediction

Himanshu Choudhary
Eindhoven University of Technology
Eindhoven, Netherlands
himanshu.dce12@gmail.com

Marwan Hassani
Eindhoven University of Technology
Eindhoven, Netherlands
m.hassani@tue.nl

ABSTRACT

In urban landscapes, traffic congestion, often identified by outlier events like accidents or constructions, poses a significant challenge. These outliers result in abrupt traffic fluctuations, necessitating real-time modeling for accurate traffic predictions. The proposed Outlier Weighted Autoencoder Modeling (OWAM) framework addresses this by employing autoencoders for local outlier detection at each traffic sensor and generating correlation scores to assess neighboring traffic's impact. These scores, which serve as the weighted information of the neighboring sensors, enhance the model's performances and enable effective real-time updates. OWAM achieves a balance between accuracy and efficiency, making it highly suitable for real-world applications. This advancement in traffic prediction models significantly contributes to the field of transportation management. The framework and its datasets are publicly available under https://github.com/himanshudce/OWAM.

ACM Reference Format:
Himanshu Choudhary and Marwan Hassani. 2024. Autoencoder-based Continual Outlier Correlation Detection for Real-Time Traffic Flow Prediction. In *The 39th ACM/SIGAPP Symposium on Applied Computing (SAC '24), April 8–12, 2024, Avila, Spain.* ACM, New York, NY, USA, Article 4, 3 pages. https://doi.org/10.1145/3605098.3636162

1 INTRODUCTION

Traffic congestion, a major urban challenge, adversely impacts public health, economic stability, and environmental quality due to increased emissions and air pollution [18]. Optimizing existing traffic systems is crucial for enhancing traffic flow, as expanding road capacity is not always feasible. While traffic modeling has seen significant advancements, challenges persist, particularly in dealing with outlier scenarios such as accidents, events, or roadwork, where traffic patterns deviate markedly from the norm. These situations are sudden and can completely alter the flow of traffic, therefore, demand real-time traffic modeling solutions that are capable of accommodating such outlier and dynamic shifts in traffic concurrently.

Our study introduces the Outlier Weighted Autoencoder Modeling (OWAM) framework, designed to improve traffic predictions in both dynamic and outlier scenarios. Leveraging Autoencoders (AE) for real-time local outlier detection, OWAM integrates Earth Mover's distance (EMD)[1] as a loss function in AE to enhance performance. Based on the outlier score, we calculate correlations between the target sensor and its neighbors and, subsequently, merge the neighboring data with the target data, assigning weights based on their correlation scores to ensure precise influence. Furthermore, we incorporate outlier-based model updates through OWAM to effectively address both outlier scenarios and the dynamic changes simultaneously. We evaluate the framework using three different real-world datasets: Hague, METR-LA, and PEMS-BAY. The key contributions of this research are the innovative OWAM framework for enhanced traffic flow prediction, the novel outlier detection architecture integrating EMD loss, the extensive evaluation of real-world datasets, and the provision of an open-source implementation for further research. An extensive exploration of OWAM with a deeper experimental evluation is available in [6].

2 BACKGROUND AND RELATED WORK

In the field of outlier detection and traffic modeling, traditional methods like One-Class SVMs, Local Outlier Factors (LOF), and Isolation Forests have been used to identify data points deviating from normal patterns [4, 14, 20]. However, these techniques often fall short in real-time applications like traffic flow forecasting due to scalability issues and high response times. Autoencoders (AEs) have emerged as a more effective approach, particularly in online scenarios (with algorithms like KitNET [17] and MemStream [2]). Probability Weighted Autoencoders (PW-AE) [5] further enhance the AE approach by optimizing loss functions based on anomaly probability, efficiently handling concept drift and data stream anomalies.

Traffic modeling on the other hand has evolved to include parametric, machine learning, and deep learning techniques [15]. Hybrid and deep learning methods like CNNs and LSTMs, that can extract space-time dependencies have shown an improved performance [11, 13]. Moreover, the use of Flow Probability Distributions (FPDs) and Earth Mover's distance has shown promising results in effectively detecting outliers in traffic data [1, 7, 8, 16].

However, these approaches often lack in handling the real-time updates and outlier scenarios simultaneously, which is necessary to capture the dynamic nature of traffic. This creates a research gap, as there is a need for models that can manage both outlier scenarios and real-time adjustments efficiently.

3 METHODOLOGY

The Outlier Weighted Autoencoder Modeling (OWAM) framework, illustrated in Figure 2, offers a novel approach to traffic modeling. It begins by aggregating traffic data at 5-minute intervals and creating Flow Probability Distributions (FPDs) of every 1-hour window

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SAC '24, April 8–12, 2024, Avila, Spain
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0243-3/24/04.
https://doi.org/10.1145/3605098.3636162

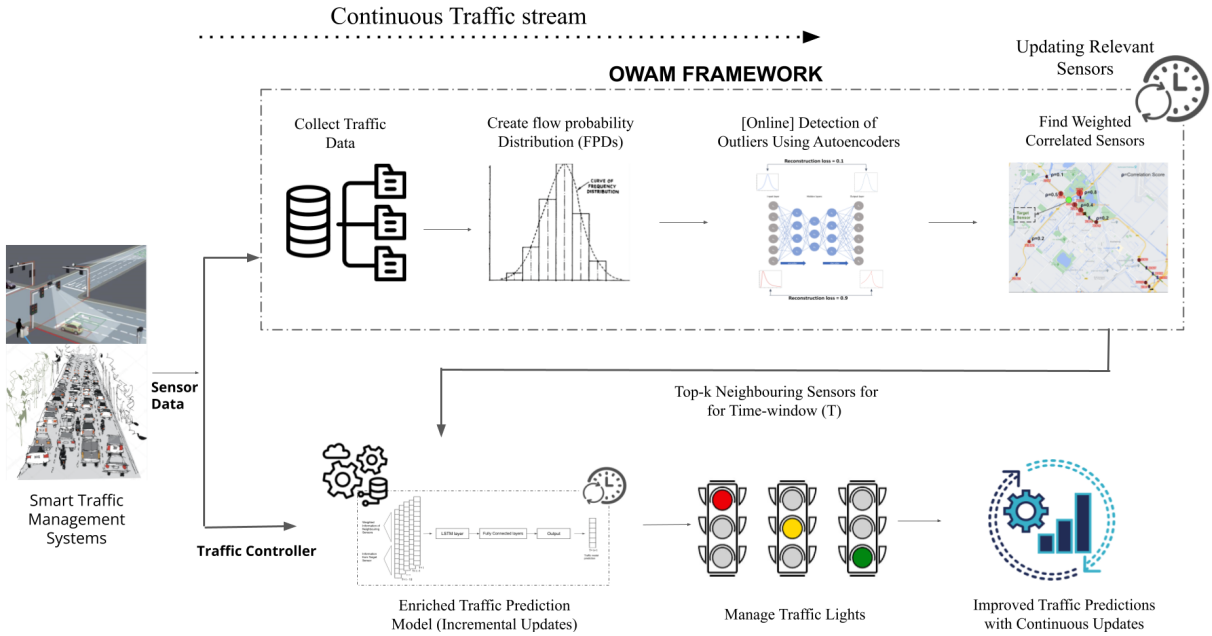


Figure 1: Overview of OWAM Framework with Real-Time Updates

to capture high-level traffic trends and reduce noise. Very short intervals for FPDs can introduce excessive noise, while very long intervals may over smooth data, potentially missing important patterns.

In the core phase of outlier detection, OWAM employs online autoencoders (AE) for real-time analysis of FPDs, identifying local anomalies at each sensor. A novel aspect of this process is the integration of Earth Mover's distance (EMD) loss with AE, improving the accuracy of outlier detection. The framework retains the raw outlier scores for each sensor, ensuring a comprehensive understanding of traffic dynamics without the need for additional threshold filtering.

OWAM then calculates correlation scores between the target sensor and its neighbors using the outlier scores. These scores serve as weights to adjust the significance of neighboring sensor information before its integration with the target sensor's data. This weighted data is concatenated with target sensor data and fed into an LSTM model for efficient traffic prediction. The methodology highlights that OWAM can enhance not just LSTM but any traffic prediction model.

The framework is designed to adapt to the dynamic nature of traffic through an incremental updating strategy. It recalculates outlier and correlation information periodically for each time window and incrementally updates the LSTM model with this new information. Starting with historical data, the model integrates data from subsequent time windows using a test-then-train methodology [3], ensuring real-time adaptability.

To evaluate and analyze the model, our research utilizes three real-world datasets, each capturing different aspects of traffic flow as outlined in Table 1.

Table 1: Dataset Information

Dataset	Hague	METR-LA	PEMS-BAY
#Samples	236,736	34,272	52,116
#Sensors	23	207	325
Sample Rate	5 minutes	5 minutes	5 minutes
Time Range	2 Years	4 months	6 months
Traffic Indicator	#Vehicles	Speed (mi/h)	Speed (mi/h)

4 RESULTS AND CONCLUSION

To evaluate we first investigated OWAM's impact on the dimensionality of the traffic model, finding that the optimal number of sensors for inclusion varies by dataset. The Hague dataset, due to the lower number of sensors, favored the inclusion of all neighboring sensors, while the METR-LA and PEMS-BAY datasets required a more selective approach, favoring the top 10% to 50% of neighboring sensors for improved predictions. It's worth noting that utilizing only the target sensor or incorporating all available sensor information of the traffic network results in lower prediction performance. An increase in training time is noticeable for the higher-dimensional PEMS-BAY dataset w.r.t. others.

The integration of Earth Mover's Distance (EMD) loss in Autoencoders within OWAM resulted in superior performance, demonstrating a 5% to 11% improvement for the METR-LA and PEMS-BAY datasets as compared to models using RMSE loss. The use of EMD loss led to a higher variance in outlier scores, enabling the model to better differentiate between more and less relevant sensors.

Comparatively, OWAM outperformed the LSTM Baseline and previous outlier-based models in terms of RMSE scores, as shown in Table 2. While matching the state-of-the-art DGCRN model in the Hague dataset, it was outperformed by DGCRN in the METR-LA

Table 2: Combined Results for Hague, METR-LA, and PEMS-BAY datasets

Dataset	Model Name	RMSE	Train Time	Eval Time
Hague	OWAM (Ours)	*13.479*	677.74	**47.617**
	HST[21]	13.631	**244.31**	101.32
	Kit-Net[17]	13.540	*305.37*	*48.267*
	DGCRN[12]	**13.110**	5043.31	280.00
	OBIS[16]	13.547	682.81	51.30
	LSTM Baseline	13.547	682.81	29.83
METR-LA	OWAM (Ours)	*5.880*	**46.18**	**8.09**
	HST[21]	6.815	*47.53*	27.37
	Kit-Net[17]	5.883	55.44	11.58
	DGCRN[12]	**4.161**	11263.23	310.04
	OBIS[16]	7.421	91.37	*10.78*
	LSTM Baseline	8.831	106.06	11.11
PEMS-BAY	OWAM (Ours)	*4.036*	**303.94**	**97.56**
	HST[21]	4.816	372.86	110.73
	Kit-Net[17]	4.312	*356.96*	104.393
	DGCRN[12]	**1.591**	25900.67	780.04
	OBIS[16]	4.375	377.49	*99.27*
	LSTM Baseline	5.434	405.05	111.74

and PEMS-BAY datasets due to DGCRN's superior spatio-temporal dependency handling. However, OWAM's much faster processing times make it a more practical choice for real-world applications, especially in large-scale, real-time traffic modeling where rapid data processing and decision-making are key. OWAM's ability to efficiently handle larger datasets and complex scenarios with less computational burden than DGCRN makes it a scalable, time-efficient solution for traffic management and similar applications.

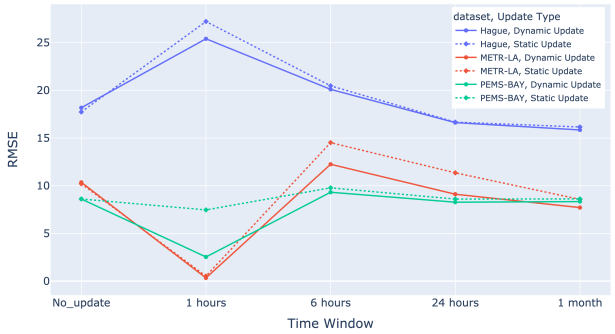


Figure 2: Overview of OWAM Framework with Real-Time Updates

In real-time settings, OWAM demonstrated its effectiveness by significantly enhancing performance through dynamic updates based on outlier-weighted correlations as indicated in Figure 2. This improvement was particularly noticeable in both short (1-hour) and longer (daily or monthly) time windows, showcasing its adaptability to various traffic scenarios. This emphasizes the necessity of selecting appropriate time windows for model updates based on data analysis and pattern study.

In conclusion, the Outlier Weighted Autoencoder Modeling framework (OWAM) presents a novel approach to outlier-driven traffic modeling, effectively merging advanced outlier detection techniques with real-time traffic analysis. By incorporating autoencoders with Earth Mover's Distance loss, OWAM significantly enhances outlier detection and integrates anomaly information into traffic prediction models, offering dynamic updates for real-time modeling. Experimental results demonstrate OWAM's superior performance, outperforming traditional models and providing an efficient solution for real-time traffic applications. In the Future, we plan to incorporate elements from GNN in our architecture to achieve an equilibrium between the superior accuracy of GNNs and the rapid processing capabilities of outlier-based models. Additionally, weight allocation and model training into one unified network through the use of an attention mechanism could yield a robust and interpretable model. The applicability of OWAM extends to fields requiring anomaly detection and real-time concept drift detection [9, 10, 19], opening new avenues for research in diverse domains.

REFERENCES

[1] Erik Andersen, Marco Chiarandini, Marwan Hassani, Stefan Jänicke, Panagiotis Tampakis, and Arthur Zimek. 2022. Evaluation of Probability Distribution Distance Metrics in Traffic Flow Outlier Detection. In *MDM*.

[2] Siddharth Bhatia, Arjit Jain, Shivin Srivastava, Kenji Kawaguchi, and Bryan Hooi. 2022. MemStream: Memory-Based Anomaly Detection in Multi-Aspect Streams with Concept Drift. (2022).

[3] Albert Bifet, Geoff Holmes, Bernhard Pfahringer, and Andreas Pfisterer. 2018. *Machine Learning for Data Streams: With Practical Examples in MOA*. MIT Press.

[4] Markus M. Breunig, Hans-Peter Kriegel, Raymond T. Ng, and Jörg Sander. 2000. LOF: Identifying Density-Based Local Outliers. (2000).

[5] Lucas Cazzonelli and Cedric Kulbach. 2023. Detecting Anomalies with Autoencoders on Data Streams. In *ECML PKDD*.

[6] Himanshu Choudhary and Marwan Hassani. 2023. Enhancing Traffic Flow Prediction using Outlier-Weighted AutoEncoders: Handling Real-Time Changes. arXiv:2312.16596 [cs.LG]

[7] Youcef Djenouri and Arthur Zimek. 2018. Outlier Detection in Urban Traffic Data. In *WIMS*. ACM.

[8] Wesley Fitters, Alfredo Cuzzocrea, and Marwan Hassani. 2021. Enhancing LSTM Prediction of Vehicle Traffic Flow Data via Outlier Correlations. In *COMPSAC*. 210–217. https://doi.org/10.1109/COMPSAC51774.2021.00039

[9] Marwan Hassani. 2019. Concept Drift Detection Of Event Streams Using An Adaptive Window. In *ECMS*. 230–239.

[10] Jesús Huete, Abdulhakim Ali Qahtan, and Marwan Hassani. 2023. PrefixCDD: Effective Online Concept Drift Detection over Event Streams using Prefix Trees. In *COMPSAC*. 328–333.

[11] Georgios N Kouziokas. 2021. Deep bidirectional and unidirectional LSTM neural networks in traffic flow forecasting from environmental factors. In *CSUM*.

[12] Fuxian Li, Jie Feng, Huan Yan, Guangyin Jin, Depeng Jin, and Yong Li. 2022. Dynamic Graph Convolutional Recurrent Network for Traffic Prediction: Benchmark and Solution. (2022).

[13] Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu. 2018. Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting. In *ICLR*.

[14] Fei Tony Liu, Kai Ting, and Zhi-Hua Zhou. 2009. Isolation Forest. In *ICDM*.

[15] Boris Medina-Salgado, Eddy Sánchez-DelaCruz, Pilar Pozos-Parra, and Javier E. Sierra. 2022. Urban traffic flow prediction techniques: A review. (2022).

[16] Tom Mertens and Marwan Hassani. 2022. Can we Learn from Outliers? Unsupervised Optimization of Intelligent Vehicle Traffic Management Systems. In *ECML PKDD*.

[17] Yisroel Mirsky, Tomer Doitshman, Yuval Elovici, and Asaf Shabtai. 2018. Kitsune: An Ensemble of Autoencoders for Online Network Intrusion Detection. (2018).

[18] C. Pasquale, I. Papamichail, C. Roncoli, S. Sacone, S. Siri, and M. Papageorgiou. 2015. Two-class freeway traffic regulation to reduce congestion and emissions via nonlinear optimal control. (2015).

[19] Erik Scharwächter, Emmanuel Müller, Jonathan Donges, Marwan Hassani, and Thomas Seidl. 2016. Detecting Change Processes in Dynamic Networks by Frequent Graph Evolution Rule Mining. In *ICDM*. 1191–1196.

[20] Bernhard Schölkopf, Robert C Williamson, Alex Smola, John Shawe-Taylor, and John Platt. 1999. Support Vector Method for Novelty Detection. In *NIPS*.

[21] Swee Chuan Tan, Kai Ming Ting, and Fei Tony Liu. 2011. Fast Anomaly Detection for Streaming Data. In *IJCAI*.